

Assignment-1

Q. No.	Question	CO	Bloom
1	Design a 4-bit binary adder using basic logic gates and verify its truth table for two sample inputs.	CO2	Apply
2	Construct a JK flip-flop based 3-bit asynchronous counter and draw its state diagram.	CO2	Apply
3	Analyze the hazard conditions present in a given combinational circuit and suggest a method to eliminate them.	CO2	Analyze
4	Analyze the setup time and hold time violations in a synchronous sequential circuit operating at 100 MHz.	CO3	Analyze
5	Analyze the timing diagram of a D flip-flop-based shift register and identify the critical path.	CO5	Analyze
6	Design a BCD to 7-segment decoder circuit using combinational logic and demonstrate its operation for digits 0 to 9.	CO2	Apply
7	Design a mod-6 synchronous counter using T flip-flops and draw the complete logic diagram.	CO2	Apply
8	Apply timing analysis concepts to compute the maximum operating clock frequency of a given sequential circuit.	CO3	Apply
9	Analyze a multiplexer-based combinational circuit designed to implement a specified Boolean function and justify the design choice.	CO2	Analyze
10	Analyze the static timing analysis report of a digital circuit to identify setup and hold time violations.	CO5	Analyze
11	Design a 4-bit magnitude comparator using combinational logic gates, draw its circuit diagram, and verify the output for all input combinations.	CO2	Apply
12	Design a sequence detector for the bit pattern 1011 using a Moore FSM and draw the complete state diagram, state table, and logic implementation.	CO2	Apply
13	Analyze the static timing analysis (STA) results for a multi-stage combinational-sequential circuit and identify all critical timing paths.	CO5	Analyze
14	Evaluate the effectiveness of HDL-based design and static timing analysis in resolving timing violations for an FPGA implementation, supported by debugging techniques.	CO5	Evaluate
15	Apply the concept of System-on-Chip (SoC) design to explain how IP cores are integrated into a single chip, using a simple example.	CO1	Apply
16	Analyze the advantages of using pre-verified IP cores over custom-designed blocks in an SoC development flow.	CO1	Analyze
17	Evaluate the performance trade-offs of an SoPC-based implementation compared to a dedicated ASIC for a given application.	CO1	Evaluate

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18	Analyze the data path, control path, and FSM design of a digital system developed using a top-down design methodology, and describe their interaction.	CO2	Analyze
19	Evaluate a real-world hardware-software co-design project, critically assessing the partitioning decisions, co-verification strategy, and final performance-power-cost outcome.	CO1	Evaluate
20	Apply the top-down design methodology and system modeling concepts to describe the stages of designing a digital system from specification to RTL.	CO2	Apply